

Intermediate Macro

Chapter 17: Openness in Goods and Financial Markets

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17.3 Conclusions and a Look Ahead



Key Tools and Concepts

- The **nominal exchange rate** is the foreign currency price of domestic currency.
- The **real exchange rate** is the relative price of domestic goods. An increase in either of these variables is an appreciation from the perspective of the domestic country.
- The **balance of payments** is a record of one country's transactions with the rest of the world over a given period of time. The balance of payments consists of a **current account**, which records transactions in goods and services, and a **financial account**, which records transactions in assets. Until recently this account was called the **capital account** so you still see the term used in the many countries and the press. Chapter 17 introduces basic balance of payments accounting and the balance of payments identity, which states that the current account and the capital account sum to zero.

Assumptions

- The **uncovered interest parity** condition equates the expected domestic currency returns on domestic and foreign bonds. Absent transactions costs and assuming that investors do not care about currency risk, investors will not be willing to hold both domestic and foreign bonds unless uncovered interest parity holds.

Assumptions

- We assume that domestic residents do not use foreign currency to purchase goods. This assumption is maintained through the formal work on the open economy.
- The uncovered interest parity condition assumes that investors care only about expected returns and not about risk. This assumption is maintained throughout the formal discussion of the open economy.

The Choice between Domestic Goods and Foreign Goods

17.1 Openness in Goods Markets

- Openness in goods markets means that domestic residents are able to buy foreign goods and sell domestic goods abroad.
- Goods sold to foreigners are called exports. Goods bought from foreigners are called imports.
- The difference between exports and imports is the trade balance.
- A negative trade balance is called a trade deficit, and a positive one a trade surplus.
- In an open economy, domestic residents make two decisions—
 - ① how much to spend and
 - ② how much to spend on domestic and foreign goods.
- The latter decision depends on the exchange rate.

Nominal Exchange Rates

17.1 Openness in Goods Markets

- We shall define the **nominal exchange rate** as *the price of the domestic currency in terms of foreign currency*, and denote it by E .
- For example, when looking at the exchange rate between the United States and the United Kingdom, E will denote the price of a dollar in terms of pounds (thus E was 0.79 £/\$ in December 2018).
- An **appreciation** of the domestic currency is an increase in the price of the domestic currency in terms of a foreign currency.
- A **depreciation** of the domestic currency is a decrease in the price of the domestic currency in terms of a foreign currency.

From Nominal to Real Exchange Rates

17.1 Openness in Goods Markets

- The real exchange rate depends on the nominal exchange rate E , the domestic price level P , and the foreign price level P^* .
- Suppose the United States, the domestic country, produces only one good; Cadillac luxury sedans.
- If a Cadillac luxury sedan sells for P dollars, its price in pound is EP .
- Note that E has units of $\text{£}/\text{\$}$, and P has units of $\text{\$/sedan}$, so EP has units of £/sedan .
- Now assume the United Kingdom, the foreign country, also produces only one good; Jaguar luxury sedans.
- One could compare the pound price P^* of Jaguar sedans produced in the United Kingdom to the pound price of Cadillac sedans produced in the United States.

The Construction of the Real Exchange Rates

17.1 Openness in Goods Markets

- This motivates the definition of the **real exchange rate** ε :

$$\varepsilon = \frac{EP}{P^*} \quad (17.1)$$



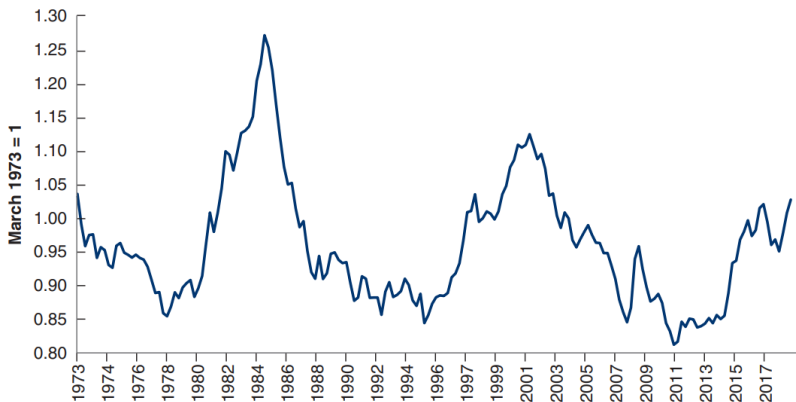
From Bilateral to Multilateral Exchange Rates

17.1 Openness in Goods Markets

- Economists are often interested in the real exchange rate measured against all other countries, rather than against just one country.
- The **bilateral exchange rates** defined in the previous slide is often replaced by **multilateral exchange rates**, which is a weighted average of the real exchange rates against all other countries.
- The multilateral exchange rates are sometimes called the **trade-weighted real exchange rates**, or the **effective real exchange rates**.
- The weight on a given country reflects two factors:
 - ① the degree to which the country trades with the domestic country and
 - ② the degree to which the country competes with the domestic country in international markets.

The US Multilateral Real Exchange Rate, since 1973

17.1 Openness in Goods Markets



Trade in Assets

17.2 Openness in Financial Markets

- Openness in financial markets means that domestic residents are able to exchange assets (bonds, stocks, and money) with residents of other countries.
- There is a link between trade in assets and trade in goods.
- Trade in assets allows countries to borrow from one another.
- Thus, countries that run trade deficits can finance them by borrowing from countries that run trade surpluses.

The Balance of Payments

17.2 Openness in Financial Markets

- The **balance of payments** summarizes the transactions of one country with the rest of the world.
- It has two components:
 - ① The first, the **current account**, is the sum of the trade balance, net investment income received from abroad, and transfers. As such, the current account is a record of net income received from the rest of the world.
 - ② The second component of the balance of payments, the **financial account**, measures the purchase and sale of foreign assets. The financial account is defined as the net decrease in foreign assets (i.e., the increase in domestic assets held by foreigners minus the increase in foreign assets held by domestic country residents).
- Apart from a statistical discrepancy, the current account and the financial account sum to zero by construction.

The US Balance of Payments, 2018, in Billions of US Dollars

17.2 Openness in Financial Markets

Table 17-3 The US Balance of Payments, 2018, in Billions of US Dollars		
Current Account		
Exports	2,500	
Imports	3,122	
Trade balance (deficit = minus sign) (1)		- 622
Income received	1,200	
Income paid	1,067	
Net income (2)		133
Current account balance (1) + (2) (deficit = minus sign)		- 489
Financial Account		
Net capital transfers (3)	9	
Increase in foreign holdings of US assets (4)	811	
Increase in US holdings of foreign assets (5)	301	
Financial account balance (7) = (3) + (4) - (5)		519
Statistical discrepancy: financial account - current account balance		30
Source: US Bureau of Economic Analysis, US International Transactions, Table. 17.1.		

Intuition behind of Balance of Payments

17.2 Openness in Financial Markets

- Think of a country as a single person.
- A country with a negative current account balance (a deficit) spends more than its income.
- To finance a deficit, it can either sell some of its existing assets to foreigners or borrow from foreigners (sell bonds to foreigner).
- By definition, these transactions have a positive sign in the financial account.
- Likewise, a country with a positive current account balance (a surplus) spends less than its income.
- It can dispose of the extra income by purchasing foreign assets or making loans to foreigners (buying foreign bonds).
- By definition, these transactions have a negative sign in the financial account.

The Choice between Domestic and Foreign Assets

17.2 Openness in Financial Markets

- Individual portfolio investment decisions are governed by the relative returns on domestic and foreign assets.
- How does one choose between domestic and foreign bonds?
- Suppose a U.S. resident has a dollar to invest.
- Let i_t be the interest rate on U.S. bonds and i_t^* the interest rate on UK bonds.
- Consider the choice between US and UK bonds

① Option 1: Buy US bonds

The return on one dollar equals $(1 + i_t)$ dollars.

② Option 2: Buy UK bonds

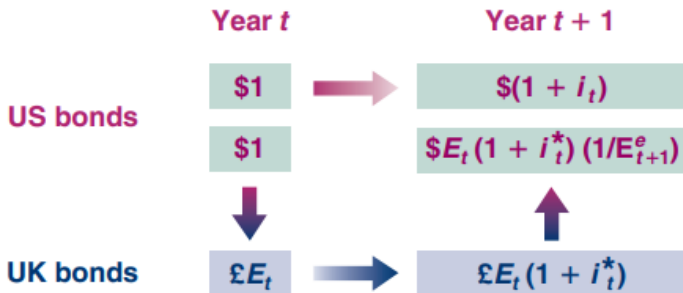
Exchange one dollar for E_t pounds.

Invest E_t pounds in UK bonds, with a return of $(1 + i_t^*)E_t$ pounds.

Exchange $(1 + i_t^*)E_t$ pounds for $\frac{(1 + i_t^*)E_t}{E_{t+1}^e}$ dollars.

Expected Returns from Holding One-Year US Bonds versus One-Year UK Bonds

17.2 Openness in Financial Markets



Interest Parity Condition

17.2 Openness in Financial Markets

- If investors care only about expected returns and not about risk, then they will choose the option with the higher expected return.
- If both US and UK bonds are to be held by the private sector, it must be that the expected returns are the same under either option.
- Arbitrage implies that the following relation must hold:

$$(1 + i_t) = (1 + i_t^*) \frac{E_t}{E_{t+1}^e} \quad (17.2)$$

- Equation (17.2) is called the **uncovered interest parity relation** or simply the **interest parity condition**.

Interest Rates and Exchange Rates

17.2 Openness in Financial Markets

- Let us get a better sense of what the interest parity condition implies. First, rewrite E_t/E_{t+1}^e as $1/[1 + (E_{t+1}^e - E_t)/E_t]$.
- Replacing in equation (17.2) gives

$$(1 + i_t) = \frac{1 + i_t^*}{[1 + (E_{t+1}^e - E_t)/E_t]} \quad (17.3)$$

which can be approximated by

$$i_t \approx i_t^* - \frac{E_{t+1}^e - E_t}{E_t} \quad (17.4)$$

- In words, equation (17.4) says that the domestic interest rate equals (approximately) the foreign interest rate minus expected appreciation of the domestic currency.
- Expected appreciation of the domestic currency makes domestic assets more attractive, so investors are willing to hold them for less compensation (a smaller interest rate).

Stage for the Study of the Open Economy

17.3 Conclusions and a Look Ahead

- Openness allows domestic residents two choices: the choice between domestic and foreign goods and the choice between domestic and foreign assets.
- Chapter 18 integrates the choice between domestic and foreign goods into the goods market equilibrium condition.
- Chapter 19 integrates the choice between domestic and foreign assets into the financial market equilibrium condition.
- Chapter 19 also combines goods and financial market equilibrium to analyze the short-run equilibrium of an open economy.
- Chapter 20 considers a medium-run model of an open economy with a fixed exchange rate, explores exchange rate crises and the behavior of flexible exchange rates, and discusses the choice of exchange rate regime.